

DMA Issues

There are lesser DMA issues with Linux these days since most of the kernels allow you to enable them without any problem, unless you are using a really old motherboard. Some of the older chipsets have faulty DMA implementation, which means DMA won't work no matter how hard you try. Another problem might be due to broken cables. In that case, replacing the cable might help.

Another area where you could have problems with DMA is when you are using devices that use the ISA bus. ISA devices do not use the standard DMA channels. So if you think DMA is creating problems, try removing the ISA cards one by one and running Linux. This will help you troubleshoot the device that's causing the problem. Upgrading your kernel might help here.

HOT Tip
rcidedma
status command helps you check if DMA is working or not.

parallel port, SCSI, and USB scanners are supported by SANE.

If your scanner is in the list, download and install SANE from <ftp://ftp.mostang.com/pub/sane>. Read the README file after unpacking the tar archive.

Using an image manipulation program like GIMP which has GTK libraries is necessary to use SANE.

After SANE has been installed, use this command to check scanner status:

scanimage -list-devices

This will give you a message like:

device

```
Load maturity level options --->
loadable module support --->
processor type and features --->
general setup --->
Memory Technology Devices (MTD) --->
parallel port support --->
Plug and Play configuration --->
lock devices --->
Multi-device support (RAID and LVM) --->
Networking options --->
telephony Support --->
ATA/IDE/MFM/RLL support --->
CSI support --->
2D device support --->
Network device support --->
 Amateur Radio support --->
rDNA (infrared) support --->
SDN subsystem --->
old CD-ROM drivers (not SCSI, not IDE) --->
input core support --->
character devices --->
Multimedia devices --->
file systems --->
console drivers --->
sound --->
SB support --->
kernel hacking --->
---
Load an Alternate Configuration File
Save Configuration to an Alternate
```

Menuconfig helps you easily load modules

#####:/dev/scanner' is a *** scanner
device 'pnm:0' is a Noname PNM file reader virtual device
device 'pnm:1' is a Noname PNM file reader virtual device**

is the name of your scanner
and ********* is the type of the scanner.

The second and third lines 'pnm:0' and 'pnm:1' will be displayed even if you don't have a scanner connected to your PC.

Now that you've got a finger on Tux's pulse and know what ails it, you can work towards quick cures to get it to work happily with different devices 

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Need more information about loading modules? Have a look at www.linux-doc.org/HOWTO/HOWTO-INDEX/index.html